

Decentralized Fake Product Detection System using Blockchain

B.Tech Project Phase 1 Presentation

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Agenda

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 - The Enterprise Backbone – Hyperledger Fabric
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Section 1: Introduction & Background

1.1 General Background

- **Global Economic Impact:** Counterfeit products cost manufacturers billions of dollars annually and pose severe safety, legal and financial risks to consumers across sectors.
- **Limitations of Traditional Measures:** Conventional anti-counterfeiting tools (printed holograms, static serial numbers, barcodes) are easily forged, copied or bypassed.
- **Lack of End-to-End Visibility:** Existing systems offer no reliable method for end consumers to trace a product's true journey from manufacturer to retail shelf.
- **Need for Decentralized Provenance:** High-value industries, including pharmaceuticals, luxury goods, and electronics, require an immutable, tamper-proof tracking infrastructure.

1.2 Problem Domain & Overview

Problem Domain

Blockchain Technology / Supply Chain Management

- **Proposed Solution:** A permissioned blockchain-based system where genuine products are registered at manufacture and tracked across every middleman handoff.
- **Consumer Verification:** End users instantly verify provenance by scanning packaging QR codes using a mobile interface connected to the ledger.
- **Keywords:** Blockchain, Hyperledger Fabric, Supply Chain Traceability, QR Code Verification, Counterfeit Detection, Decentralized Application.

Section 2: Objectives of the Project

2.1 Objectives of the Project

- **Enterprise Permissioned Ledger:** Deploy a permissioned blockchain using **Hyperledger Fabric** to record every supply chain handoff as an immutable transaction.
- **Physical-to-Digital Linking:** Implement secure QR code generation and scanning libraries to bind physical product packaging to on-chain ledger data.
- **Consumer Mobile Interface:** Build a user-friendly **React Native** mobile application for instant visual verification of product authenticity and supply chain history.
- **Efficiency & Latency Reduction:** Achieve measurable reductions in product verification time compared to traditional manual audit checks.
- **Multi-Industry Adaptability:** Design a scalable framework suited for pharmaceuticals, luxury goods, electronics, and e-commerce platforms.

Section 3: Literature Survey

Literature Survey: Presentation Order

We present the 10 reviewed papers across **five thematic phases**, moving from early approaches toward system-level evaluation:

- ① **Context, Early Approaches & Physical Tag Vulnerabilities**
- ② **Securing the Consumer-Facing Verification Layer**
- ③ **The Enterprise Backbone – Hyperledger Fabric**
- ④ **Privacy, Data Isolation & Participant Incentives**
- ⑤ **System Benchmarking, Scalability & Network Resilience**

Each phase builds toward the architectural choices made in our own system.

Phase 1: Context, Early Approaches & Physical Tag Vulnerabilities

- Establishes the baseline: how early blockchain systems approached anti-counterfeiting before Fabric-specific designs matured.
- Surfaces a key limitation our system avoids – dependency on specialized physical hardware (NFC tags) at the point of verification.

Title: Blockchain Enabled Traceability in the Jewel Supply Chain

Authors: Aman Patel, Siva Sai, Ankit Daiya, Harshal Akolekar, Vinay Chamola

Methodology

Uses Ethereum Solidity smart contracts for a 6-tier supply chain flow (Admin, Miner, Refiner, Manufacturer, Retailer, Customer). Combines on-chain storage (metadata, identifiers, Keccak-256 hashes) with off-chain IPFS dynamic certificate templates, integrated via MetaMask and React.js.

Advantages

- **Fully Decentralized:** RBAC and multisig eliminate single points of failure.
- **Scalable & Gas-Efficient:** Stores static templates on IPFS and lightweight hashes on-chain.
- **Robust Security:** Protects against reentrancy, integer errors, and DoS attacks.

Disadvantages

- **Execution Latency:** On-chain write operations take 15–18 s on testnets.
- **Adoption Barriers:** High technical training curve for non-technical actors.
- **Regulatory Conflicts:** Immutability creates tension with data privacy deletion rights.

Inference & Takeaway

Provides an operational, cost-optimized, and secure decentralized traceability framework for high-value asset tracking. However, practical industry adoption hinges on overcoming non-technical user barriers and aligning with multi-jurisdictional legal privacy requirements.

Title: Block-Supply Chain: A New Anti-Counterfeiting Supply Chain Using NFC and Blockchain

Authors: Naif Alzahrani, Nirupama Bulusu

Methodology

Integrates tamper-resistant NFC tags (storing signed product data and read counters) with a decentralized blockchain. Introduces a lightweight BFT consensus protocol based on Tendermint that replaces fixed validators with a two-stage randomized selection mechanism, assigning $\log n$ validators per block proposal.

Advantages

- **High Scalability:** Reduces communication overhead to $O(n \log n)$ by selecting $\log n$ validators.
- **Strong Detection:** Achieves up to 98.4% detection rate against tag cloning and modification.
- **Dynamic Security:** Eliminates fixed-validator bottlenecks via randomized leader-validator mapping.

Disadvantages

- **Leader Bias Risk:** Malicious validation-leaders could potentially attempt biased validator selections.
- **Fixed Validator Overhead:** Using a static $\log n$ size creates unnecessary overhead in low-hostility networks.
- **Hardware Dependence:** Relies strictly on physical NFC tags and hardware read counters.

Inference & Takeaway

Delivers an efficient, decentralized anti-counterfeiting framework for physical goods by combining NFC hardware verification with a randomized log n consensus model. It successfully balances high security with significantly lower validation latency compared to standard BFT protocols.

Phase 2: Securing the Consumer-Facing Verification Layer

- Focuses on the point where the end consumer actually verifies a product – the layer our React Native app operates at.
- Compares a QR-hardening approach against a pure computer-vision approach, informing our choice of QR + ledger verification over image classification alone.

Title: A Texture-Hidden Anti-Counterfeiting QR Code and Authentication Method

Authors: Tianyu Wang, Hong Zheng, Changhui You, Jianping Ju

Methodology

Fuses a refined QR code (reducing data modules to micro code points) with Gaussian random halftone textures to visually embed code points into a physically non-replicable pattern. Employs a DFT-based quality assessment algorithm (MAF) to screen blurred images, followed by a Dual Feature Detection Algorithm (DFDA) combining decodability checking and DFT spectral feature metrics (NC and PR).

Advantages

- **Perfect Discrimination:** DFDA achieves 100% accuracy, precision, and recall against equal and scaled replica forgeries.
- **Robust Blur Filtering:** MAF algorithm identifies defocus vs. motion blur with up to 96% accuracy to prevent false verifications.
- **High Crop/Scale Tolerance:** Preserves 100% detection accuracy under size reductions down to 48% and cut attacks up to 5%.

Disadvantages

- **Performance Drop on Heavy Cut:** Detection accuracy degrades significantly when physical cut/crop attacks exceed 5%.
- **Processing Latency:** Spectral feature calculation requires 0.25 s per code, which is slower than ultra-lightweight 2D barcode methods.
- **Lighting Sensitivity:** Relies on standard indoor capture conditions; extreme lighting or severe motion blur necessitates re-shooting.

Inference & Takeaway

Demonstrates that integrating frequency-domain spectral analysis (DFT) with error-diffused Gaussian micro-textures effectively converts standard, easily copiable QR codes into unclonable anti-counterfeiting identifiers. It solves copy vulnerability on commodity smartphones without requiring specialized optical verification hardware.

Title: Deep Neural Network-Based Detection of Counterfeit Products from Smartphone Images

Authors: Hugo Garcia-Cotte, Dorra Mellouli, Abdul Rehman, Li Wang, David G. Stork

Methodology

Employs a purely computer-vision-based system using standard smartphone images without requiring physical security tags or product modifications. Utilizes EfficientDet-D0 for emblem localization and a Spatial Transformer Network (STN) for spatial alignment invariance. Query images are classified as genuine or counterfeit via a 380-layer custom CNN architecture (AntiCounterfeit) executing cloud-based inferences.

Advantages

- **High Classification Accuracy:** Achieves 99.71% accuracy on branded garment emblems under weakly controlled real-world conditions.
- **Zero Physical Tagging:** Operates on existing products without special optical tags, physical modifications, or custom hardware.
- **Cloud Scalability:** Enables instant smartphone-based verification scalable to millions of authentication queries per day.

Disadvantages

- **Rejection Rate Trade-off:** Peak accuracy relies on rejecting 3.06% of ambiguous or poorly captured images.
- **Dataset Dependency:** Demands thousands of labeled genuine and fake image samples per product category for transfer learning.
- **Arms-Race Vulnerability:** Evolving "super fakes" necessitate continuous dataset updates and model retraining.

Inference & Takeaway

Demonstrates that deep neural networks can differentiate subtle manufacturing noise from counterfeit variations using standard smartphone cameras alone. A specialized 380-layer CNN outperforms large vision transformers (ViT-L) using one-third of the parameter count, enabling accurate, non-intrusive mobile anti-counterfeiting.

Phase 3: The Enterprise Backbone – Hyperledger Fabric

- Three permissioned-ledger systems, all built on Hyperledger Fabric – directly validating our own backbone choice.
- Spans pharma, luxury goods, and IoT-integrated tracking, showing Fabric's adaptability across the industries our abstract targets.

Title: Blockchain-based smart tracking and tracing platform for drug supply chain

Authors: Xinlai Liu, Ali Vatankhah Barenji, Zhi Li, Benoit Montreuil, George Q. Huang

Methodology

Proposes a unified five-layer Blockchain and Internet of Things (BloT³) platform built on Hyperledger Fabric using a Practical Byzantine Fault Tolerance (pBFT) consensus mechanism. Combines item-level QR codes with multi-tier RFID tracking across box, pallet, and truck levels for fine-grained identity management. Implements an on-chain/off-chain data strategy storing metadata and cryptographic hashes on-chain while keeping large raw datasets off-chain, supported by smart contracts for quality regulation, traceability, and automated risk alerts.

Advantages

- **Privacy & Integrity Balance:** Stores data hashes on-chain while keeping sensitive process data off-chain, protecting trade secrets without compromising auditability.
- **Multi-Tier Traceability:** Integrates QR codes and multi-level RFID tags to track drug identities seamlessly from item production to logistics truck levels.
- **Automated Risk Alerts:** Employs smart contract services to detect drug anomalies, counterfeit transfusions, and unauthorized data tampering in real time.

Disadvantages

- **Transaction Size Bottleneck:** System performance degrades significantly when transaction payload sizes exceed 200 kB, leading to high latency variability.
- **High Infrastructure Deployment:** Requires significant hardware setup (RFID readers, IoT sensors, gateways) across all participant facilities.
- **Stakeholder Adoption Barrier:** Demands extensive cross-organizational cooperation, data standardization, and complex user onboarding.

Inference & Takeaway

Demonstrates that integrating Hyperledger Fabric consortium blockchain with multi-tiered IoT assets provides a decentralized, tamper-proof tracking system for pharmaceutical supply chains. Empirical evaluation reveals an optimal transaction size threshold of 200 kB (achieving an efficient 792.1 ms response time), proving that hybrid on-chain/off-chain architectures can achieve real-time auditing and counterfeit prevention without compromising enterprise data privacy.

Title: PharmaChain: Blockchain-based drug supply chain provenance verification system

Authors: Sarmistha Sarna Gomasta, Aditi Dhali, Tahlil Tahlil, Md. Musfique Anwar, A.B. M Shawkat Ali

Methodology

Proposes PharmaChain, a Hyperledger Fabric-based permissioned framework using Practical Byzantine Fault Tolerance (PBFT) consensus to trace pharmaceutical distribution. Combines on-chain transaction logging with off-chain identity repositories, leveraging CouchDB for state storage and role-based read-write access control matrices. Implements a multi-layered security mechanism featuring SHA-256 hashing, Elliptic Curve Digital Signature Algorithm (ECDSA secp256k1) double signing by pharmaceutical companies and drug regulatory authorities, and a three-part governance model (Network Membership, Technical Infrastructure, and Business Network) for smart contract automation.

Advantages

- **Dual Cryptographic Authentication:** Integrates SHA-256 with dual-key ECDSA digital signatures (company and regulator) to guarantee non-repudiation and data integrity.
- **Structured Three-Tier Governance:** Defines clear operational rules across membership, technical, and business domains to harmonize cross-organizational compliance.
- **Hybrid Storage Efficiency:** Utilizes CouchDB and on-chain/off-chain access matrices to reduce network traffic while enforcing strict data privacy controls.

Disadvantages

- **Consensus Node Vulnerability:** System integrity and transaction control are vulnerable if an adversarial coalition compromises more than 33% of the network nodes.
- **Absence of Localized Ledger Replication:** Lacks support for case-specific, localized ledgers replicated across multiple geographical sites.
- **Privacy Mining Overhead:** Preserving individual privacy during research data mining requires additional mechanisms (e.g., k-anonymity) that increase computational overhead.

Inference & Takeaway

Demonstrates that pairing Hyperledger Fabric with dual-key ECDSA cryptography and multi-tier governance yields an accountable, tamper-proof drug supply chain system. Empirical evaluation shows that hybrid on-chain/off-chain CouchDB architectures combined with smart contract-based automated alerts enable real-time regulatory oversight, verifiable provenance tracking, and effective anti-counterfeiting without exposing sensitive operational data.

Title: Blockchain-Based Anti-Counterfeiting Management System for Traceable Luxury Products

Authors: Chin-Ling Chen, Long-Hui Guo, Ming Zhou, Woei-Jiunn Tsaur, Hongyu Sun, Wanbing Zhan, Yong-Yuan Deng, Chun-Ta Li

Methodology

Proposes a Hyperledger Fabric consortium blockchain framework to prevent data forgery and enable anti-counterfeiting traceability across the luxury product supply chain. Integrates smart contracts (chaincode) with the Elliptic Curve Digital Signature Algorithm (ECDSA) to ensure data integrity, non-repudiation, and transparent tracking from raw material sourcing to consumer purchase. Features asymmetric public-key encryption and timestamp validation to prevent Man-in-the-Middle (MitM) and replay attacks during network communication. Incorporates a multi-role structure (brand, manufacturer, supplier, distributor, logistics provider, deliverymen, consumers, and third-party verifiers) with dual-confirmation scanning and signing protocols during logistics handovers.

Advantages

- **End-to-End Multi-Role Traceability:** Connects all supply chain entities into a consortium network for complete lifecycle product tracking.
- **Comprehensive Cryptographic Defense:** Integrates ECDSA signatures, asymmetric encryption, and timestamps to prevent replay and Man-in-the-Middle attacks.
- **Verified Logistics Handovers:** Enforces dual-confirmation code scanning and cryptographic signatures between deliverymen and recipients at each transfer point.

Disadvantages

- **High Operational Complexity:** Requires active key management and cryptographic signing operations from delivery personnel and end consumers.
- **Physical-Digital Mapping Risk:** Cannot independently prevent physical item swapping prior to digital barcode or QR code scan verification.
- **Consensus & Storage Overhead:** Frequent multi-party smart contract triggers and transaction logging across detailed logistics transfers increase network overhead.

Inference & Takeaway

Demonstrates that pairing Hyperledger Fabric with ECDSA digital signatures and multi-party smart contracts establishes a highly accountable, tamper-proof anti-counterfeiting framework for luxury goods. Requiring dual-party scan-and-sign confirmation during logistics handovers ensures transparent chain-of-custody verification while keeping transactional communications secure and verifiable without relying on a centralized authority.

Phase 4: Privacy, Data Isolation & Participant Incentives

- Addresses a problem Fabric alone doesn't solve: keeping competitively sensitive supply-chain data private between organizations.
- Introduces incentive design as a way to keep participant-submitted data trustworthy before it ever reaches the ledger.

Title: Design and Implementation of a Trusted Food Supply Chain Traceability System with Incentive Using Hyperledger Fabric

Authors: Zhiyang Zhou, Yaokai Feng, Kouichi Sakurai

Methodology

Proposes a Hyperledger Fabric multi-channel architecture to partition sensitive merchant business data into private sub-channels while using a main channel for cross-organizational traceability. Implements dynamic channel-level Access Control Lists (ACLs) via smart contracts (chaincode) to regulate cross-channel data queries. Incorporates a government regulator node to supervise enterprises and introduces a dynamic trust evaluation model combining buyer ratings with regulatory scores. Includes a dynamic reward-and-penalty mechanism that updates merchant composite scores, assigns trust grades, and automatically revokes transaction permissions for banned entities to mitigate the "Garbage In, Garbage Out" (GIGO) problem prior to on-chain recording.

Advantages

- **Privacy-Preserving Multi-Channel Structure:** Isolates sensitive commercial data in dedicated sub-channels with strict ACL access control.
- **GIGO Mitigation via Dual-Scoring:** Mitigates data falsification by combining buyer reviews and government regulatory scores into dynamic trust updates.
- **Lightweight & Stable Incentive Mechanism:** Provides automated grading and banning with minimal computational overhead compared to heavy mathematical models.

Disadvantages

- **Lack of Endpoint Physical Verification:** Does not feature a "one item, one certificate" binding (e.g., via NFTs), allowing counterfeit goods to share traceability records.
- **On-Chain Storage Overhead:** Storing full traceability records directly on-chain increases ledger size and creates potential storage bottlenecks.
- **Throughput Degradation Under High Concurrency:** Performance tests reveal significant latency spikes and lower throughput when client worker threads exceed optimal thresholds (e.g., 50 workers).

Inference & Takeaway

Demonstrates that mitigating the GIGO problem requires addressing data credibility before on-chain storage rather than relying solely on blockchain immutability. Combining multi-channel privacy isolation with a lightweight dual-rating incentive framework (buyers + regulators) effectively promotes truthful data submission while maintaining stable execution performance in food supply chain networks.

Phase 5: System Benchmarking, Scalability & Network Resilience

- Moves from architecture to evaluation – what throughput, latency, and failure rates a Fabric-based system can realistically deliver.
- Covers resilience against network-layer attacks, a concern our system must also account for once deployed.

Title: Enhancing Traceability and Reliability in Cold Chain Logistics Through Hyperledger Fabric and IoT

Authors: Elvan Duman, Ebru Aydoğan

Methodology

Proposes a permissioned blockchain infrastructure using Hyperledger Fabric integrated with IoT technology to address data centralization, lack of transparency, and cross-organizational trust in cold chain logistics. The multi-organizational architecture links producers, carriers, and retailers on a shared channel using RAFT consensus and Role-Based Access Control (RBAC) integrated via Membership Service Providers (MSPs). Low-power BLE 4.2 sensors and ESP32 gateways operating in deep-sleep mode capture ambient temperature and humidity data, transmitting them via MQTT to a REST API layer. The API authenticates clients via JWT and invokes chaincode to immutably log time-stamped environmental parameters to the ledger. Includes an unauthenticated public web interface allowing consumers to scan product QR codes and verify historical supply chain environmental metrics against predefined safe temperature bands.

Advantages

- **Low-Power IoT Design:** Efficient ESP32 deep-sleep duty cycling enables continuous cold chain monitoring for 7 to 10 days on a 2000 mAh battery.
- **High Query Performance & Reliability:** Query workloads achieve up to 818 TPS with sub-second average latency (0.26 s) and 0% transaction failure rate.
- **Simplified Multi-Org RBAC Governance:** Role-based identity management reduces operational complexity compared to ABAC while enabling transparent public QR verification.

Disadvantages

- **Lack of Predictive Intelligence:** Focuses strictly on reactive monitoring without embedding AI/ML models to forecast cold-chain disruption risks in advance.
- **Write Transaction Bottlenecks:** Complex transfer operations exhibit higher average latencies (up to 6.30 s under 200 TPS send rates) due to multi-peer endorsement overhead.
- **Limited Cryptographic Future-Proofing:** Lacks post-quantum cryptographic primitives and advanced cross-chain interoperability mechanisms.

Inference & Takeaway

Demonstrates that coupling permissioned blockchains with low-power IoT hardware provides scalable, zero-failure data immutability for cold supply chains. Substituting complex attribute-based access controls with structured Role-Based Access Control (RBAC) streamlines multi-party governance, while horizontal peer scaling effectively suppresses latency growth under high-concurrency read-write workloads.

Title: Semi-Supervised Learning for Anomaly Detection in Blockchain-based Supply Chains

Authors: Do Hai Son, Bui Duc Manh, Tran Viet Khoa, Nguyen Linh Trung, Dinh Thai Hoang, Hoang Trong Minh, Yibeltal Alem, Le Quang Minh

Methodology

Proposes a parallel semi-supervised Deep AutoEncoder-Multilayer Perceptron (DAE-MLP) framework to detect cyber-attacks across network, consensus, and application layers of Ethereum-based supply chains using network-layer traffic data. Operating at the node detector level, a traffic monitor extracts 21 features via the BC-ID tool from raw network packets generated across four supply chain roles (farmer, distributor, retailer, consumer). The model runs an unsupervised DAE (trained on normal traffic to compute reconstruction error) in parallel with a supervised 2-class MLP (predicting anomaly probabilities). Their outputs combine into a unified anomaly score evaluated against an adaptive threshold computed using a grid-search algorithm with decay rate.

Advantages

- **High Cross-Layer Detection Accuracy:** Achieves 96.5% overall accuracy, outperforming pure unsupervised models and traditional binary classifiers.
- **Efficient Retraining & Adaptability:** Allows independent hyper-parameter updates to the MLP branch for newly identified attacks without retraining the DAE.
- **Cross-Layer Visibility from Network Layer:** Detects consensus/application-layer attacks (e.g., DoS block gas limit, Overflow/Underflow, FoT) solely using network-level packet features.

Disadvantages

- **Supervised Dependency for Zero-Day Attacks:** Initial detection performance drops for un-updated novel attacks before the MLP component is retrained.
- **Threshold Tuning Sensitivity:** Detection accuracy is highly sensitive to the initial quantile threshold setting and requires re-running grid search after updates.
- **Single-Framework Testing Scope:** Evaluation relies on lab-simulated Ethereum private networks rather than large-scale production multi-chain environments.

Inference & Takeaway

Demonstrates that combining unsupervised reconstruction error with supervised classification in a parallel architecture effectively captures subtle smart-contract and consensus-layer attacks using only network-layer data streams. Modular decoupling of the supervised classifier enables dynamic adaptation to zero-day threats in decentralized supply chains without full model re-training.

Section 4: Literature Study Table

Literature Study Table (1/5)

Phase 1: Context, Early Approaches & Physical Tag Vulnerabilities

Title & Authors	Methods Used	Features	Advantages	Disadvantages	Metrics & Future Scope
Paper 1: Jewel Supply Chain Traceability (Patel et al.)	Ethereum Solidity smart contracts; 6-tier chain (Admin→Customer); Keccak-256 hashing; IPFS off-chain storage	On-chain meta-data + off-chain IPFS certificates; MetaMask/React.js integration	Fully decentralized RBAC/multisig; gas-efficient hybrid storage; robust against reentrancy/DoS	On-chain writes take 15–18 s; steep training curve for non-technical actors; immutability vs. privacy-law conflicts	Gas cost, tx latency <i>Future:</i> Cross-chain interoperability, simplified UX
Paper 2: Block-Supply Chain NFC (Alzahrani & Bulusu)	NFC tags with signed data + read counters; lightweight BFT consensus (Tendermint-based) with randomized log n validator selection	Two-stage validator randomization per block; tag-cloning resistant hardware layer	$O(n \log n)$ communication overhead; 98.4% tag-clone detection rate; no fixed-validator bottleneck	Malicious leader could bias validator selection; static log n overhead in low-risk networks; hard dependency on NFC hardware	Detection rate, consensus latency <i>Future:</i> Leader-selection hardening

Literature Study Table (2/5)

Phase 2: Securing the Consumer-Facing Verification Layer

Title & Authors	Methods Used	Features	Advantages	Disadvantages	Metrics & Future Scope
Paper 3: Texture-Hidden Anti-Counterfeiting QR (Wang et al.)	Gaussian halftone micro-texture fused into refined QR code; DFT-based MAF blur screening; Dual Feature Detection Algorithm (NC, PR spectral metrics)	Physically non-replicable visual pattern; decodability + spectral dual-check	100% accuracy/precision/recall vs. replicas; 96% blur-type discrimination; 100% accuracy under 48% size reduction	Accuracy drops sharply beyond 5% cut/crop; 0.25 s/code processing overhead; sensitive to extreme lighting	Accuracy, NC/PR scores <i>Future:</i> Faster spectral computation
Paper 4: DNN Counterfeit Detection from Smartphone Images (Garcia-Cotte et al.)	EfficientDet-D0 emblem localization; Spatial Transformer Network alignment; 380-layer custom CNN (AntiCounterfeit), cloud inference	Pure computer-vision pipeline; no physical tags or product modification needed	99.71% accuracy on real-world garment emblems; zero physical tagging required; scales to millions of daily cloud queries	Peak accuracy relies on rejecting 3.06% of ambiguous images; needs large labeled datasets per category; vulnerable to evolving "super fakes"	Accuracy, rejection rate <i>Future:</i> Continuous retraining pipelines

Literature Study Table (3/5)

Phase 3: The Enterprise Backbone – Hyperledger Fabric

Title & Authors	Methods Used	Features	Advantages	Disadvantages	Metrics & Future Scope
Paper 5: Drug Supply Chain (Liu et al.)	5-layer Hyperledger Fabric + pBFT; item-level QR + multi-tier RFID (box/pallet/truck); on-chain hashes, off-chain raw data	Smart contracts for quality regulation & automated risk alerts	Balances privacy and auditability; fine-grained multi-tier traceability; real-time anomaly alerts	Performance degrades past 200 kB payloads; heavy hardware deployment (RFID/IoT); high onboarding complexity	TPS, response time (792.1 ms @200 kB) <i>Future:</i> Payload optimization
Paper 6: PharmaChain (Gomasta et al.)	Hyperledger Fabric + PBFT; CouchDB state storage; SHA-256 + dual ECDSA (secp256k1) signing; 3-part governance model	RBAC read-write matrices; dual company+regulator signatures	Strong non-repudiation via dual crypto signing; clear cross-org governance structure; efficient hybrid CouchDB storage	Vulnerable if >33% nodes compromised; no localized/geo-replicated ledgers; k-anonymity needed for privacy-safe mining	Node fault threshold <i>Future:</i> k-anonymity integration
Paper 7: Luxury Product Traceability (Chen et al.)	Hyperledger Fabric consortium chain; ECDSA chaincode signing; asymmetric encryption + timestamps (anti-MitM/replay)	Multi-role structure (brand→consumer); dual confirmation scan-and-sign handovers	End-to-end multi-role traceability; strong MitM/replay defense; verified logistics handovers	High key-management burden on delivery staff; can't stop physical swap pre-scan; heavy multi-party contract overhead	Handover verification rate <i>Future:</i> Simplify consumer-side key mgmt

Literature Study Table (4/5)

Phase 4: Privacy, Data Isolation & Participant Incentives

Title & Authors	Methods Used	Advantages	Disadvantages	Metrics & Future Scope
Paper 8: Trusted Food Supply Chain with Incentive (Zhou, Feng, Sakurai)	Hyperledger Fabric multi-channel architecture with dynamic ACLs; government regulator node; dual buyer+regulator trust scoring; reward-penalty grading with auto-revocation	Isolates sensitive commercial data per sub-channel; mitigates GIGO problem via dual-scoring before on-chain write; lightweight, low-overhead incentive design	No item-level physical binding (e.g. NFT-style "one item, one certificate"); on-chain storage overhead grows with full records; throughput drops past ~50 concurrent workers	Throughput vs. concurrency <i>Future:</i> NFT-based item-certificate binding

Literature Study Table (5/5)

Phase 5: System Benchmarking, Scalability & Network Resilience

Title & Authors	Methods Used	Features	Advantages	Disadvantages	Metrics & Future Scope
Paper 9: Cold Chain Traceability (Duman & Aydoğan)	Hyperledger Fabric + RAFT consensus; RBAC via MSPs; BLE 4.2 + ESP32 sensors; MQTT → REST (JWT) → chain-code	Public QR interface for consumer environmental-history checks	Low-power IoT: 7–10 days on 2000 mAh; 818 TPS query throughput, 0.26 s latency, 0% tx failure; simpler RBAC than ABAC	No predictive AI for disruption forecasting; write ops up to 6.30 s @200 TPS; no post-quantum cryptography	TPS, latency, failure rate <i>Future:</i> Predictive ML, PQC readiness
Paper 10: Semi-Supervised Blockchain Anomaly Detection (Son et al.)	Parallel DAE (unsupervised) + MLP (supervised) on 21 BC-ID network features; adaptive grid-search threshold	Detects network/consensus/application-layer attacks from traffic alone	96.5% overall detection accuracy; MLP branch retrains independently for new attacks; no need for consensus-layer instrumentation	Weak on un-updated zero-day attacks; threshold highly sensitive to tuning; tested only on lab-simulated Ethereum networks	Accuracy, F1-score <i>Future:</i> Production-scale multi-chain validation

Thank You!